Which 16 of 24? Per-Die Silicon Harvest Maps, Procurement Lots and Core Guarding from Supercomputer Telemetry

Mohsen Seyedkazemi Ardebili[1] Michael Bidollahkhani[2] Andrea Bartolini[1]

[1]*Department of Electrical, Electronic and Information Engineering*, *University of Bologna*, Italy

[2]*University of Göttingen*, Germany

{mohsen.seyedkazemi, a.bartolini}@unibo.it michael.bkhani@uni-goettingen.de

***Abstract*—Processor vendors sell partially defective dies by permanently fusing off the faulty units. The resulting per-die maps of *which* units were disabled are proprietary and conventionally invisible to the buyer. We show that the out-of-band sensor telemetry supercomputers already collect discloses them: a baseboard management controller exposes one thermal sensor per physical core position and does not renumber survivors, so the set of reporting sensors *is* the harvest map. We recover maps for all 1,962 POWER9 sockets of CINECA's Marconi100 from a public dataset [1] and track them across 1.67 million socket-days.**

Harvesting operates exactly at the POWER9 *slice*—the pair of cores sharing an L2/L3 block—with zero exceptions, and is per-die rather than per-SKU: 443 of the 495 possible patterns occur. A thermal probe over all 1,948 sockets independently recovers the sensor index as the physical ordering of slices on the die ($z = -6.55$; an unconstrained search returns the same ordering reversed, $|\rho| = 0.993$), which establishes that harvested slices cluster *spatially*. A fitted pairwise-interaction model separates that clustering from per-slice propensity and shows a sharp procurement boundary at rack 22, calibrated against a permutation null and sharp enough to assign an unseen rack to the right delivery 96 % of the time: the two lots differ in both. Maps are static—changing at reboot and essentially nowhere else, which we verify against an independent boot signal rather than infer from gaps—and we isolate three field core-guard episodes. A taxonomy separates processor replacements from node replacements, socket relabelling and firmware deconfiguration, giving an annualised socket replacement rate of 2.1 % with no asset database. An equivalence analysis bounds the map's operational footprint—it moves socket power about as much as which slot a processor occupies, and predicts field replacement not at all.

***Index Terms*—core harvesting, silicon binning, HPC telemetry, POWER9, yield, hardware provenance**

I. Introduction

Every large processor die ships with defects. Rather than discard them, vendors permanently fuse off the faulty units and sell the remainder as a lower-core-count part. We call this *silicon harvesting*, or *die harvesting*, to distinguish it from the now more common architectural sense of "core harvesting" as the reclamation of idle CPU *cycles* for co-located work [2]; the two are unrelated. The practice is universal and openly acknowledged, but the resulting *per-die* maps of which units were disabled are proprietary: a buyer is told the part has 16 cores, never *which* 16. Those maps encode information about a fab's defect process, a vendor's binning policy and a procurement's provenance that is otherwise unavailable outside the manufacturer.

Independently, HPC operators have begun publishing holistic out-of-band monitoring at fleet scale [1]. We observe that such telemetry carries an unintended side channel: a baseboard management controller exposes one thermal sensor per *physical* core position and does not renumber the surviving cores compactly, so the set of sensors that report *is* the die's harvest map, published inadvertently and at fleet scale.

We exploit this to recover harvest maps for every processor socket of CINECA's Marconi100, a Top-10 system, from a public dataset, and to track them across 858 days. Our contributions are:

- A recovery method (Sec. III) with an explicit validity argument, needing only per-core sensor telemetry that operators already collect, **validated against the operating system's and the scheduler's own core counts** (Sec. III-B).
- The first fleet-scale characterisation of core harvesting in a deployed supercomputer: the granularity is exactly the POWER9 slice, maps are per-die rather than per-SKU, and harvested slices cluster in sensor-index space.
- Evidence of *procurement-lot structure*—a sharp change-point separating two populations with markedly different binning statistics—from telemetry alone.
- A longitudinal result over 1.67 M socket-days: maps are static, configuration changes coincide with reboots **measured against an independent boot signal** rather than inferred from reporting gaps (Sec. IV-H), and three field core-guard episodes are isolated and corroborated by two further collectors.
- A taxonomy that separates processor replacements, node replacements, socket relabelling and firmware deconfiguration from one another (Sec. IV-I), yielding an annualised socket replacement rate of 2.1 % with no asset database.
- Two bounded nulls that matter operationally: an equivalence analysis showing the map's effect on socket power is comparable to socket position rather than negligible (Sec. IV-L), and a survival model showing it does not predict field replacement at all (Sec. IV-M).
- A quantification of the disclosure itself (Sec. IV-N):

recovery costs about two minutes of ordinary telemetry, and reducing resolution does not mitigate it.

II. Background

A. The machine, from room to core

Marconi100's 980 compute nodes occupy 49 rack cabinets of 20 nodes each, arranged in three rows of the machine room, with $\mathrm{id} = 20\,\mathrm{rack} + \mathrm{slot}$ and every rack's (X, Y) floor position documented [3]. Each node is an AC922 8335-GTG chassis carrying *two* CPU sockets and four V100 GPUs. Each socket holds one POWER9 die; each die carries 24 SMT4 core positions organised as 12 slices; each slice is two cores plus their shared L2 and L3. Fig. 1 shows the chain. Two consequences matter throughout: the unit we recover a map for is a *socket*—two per node, not one—and the unit that gets fused is a *slice*, not a core. Sec. IV-A fixes the exact socket populations.

B. POWER9 slice organisation

The POWER9 die used in the IBM Power System AC922 contains 24 SMT4 cores (equivalently 12 SMT8 cores) on a $695\,\mathrm{mm}^2$ die in $14\,\mathrm{nm}$ FinFET with 17 metal layers [4]. Critically, *each pair of SMT4 cores—or one singleton SMT8 core—comprises a* slice*, and each slice owns 512 kB of L2 and 10 MB of L3* [4], [5]. IBM's Hot Chips 28 presentation shows the 24 cores above $12 \times 10\,\mathrm{MB}$ L3 blocks, attached to a 7 TB/s on-chip fabric at $256\,\mathrm{GB/s} \times 12$ [5]. The slice is therefore the natural fusing granularity: disabling one SMT4 core would strand its half of a shared L2/L3 block.

Marconi100 nodes are AC922 model 8335-GTG with $2\times$ 16-core POWER9 at 2.6/3.1 GHz and $4\times$ NVIDIA V100 [6], [7]. **A 16-core part is a 24-core die with four slices fused off**; this is the object of our study. Fig. 2 shows the organisation annotated with the harvest rates we measure.

C. The M100 ExaData dataset

M100 ExaData [1] publishes 49.9 TB of telemetry from Marconi100 covering 2020-03 to 2022-09 across 980+ nodes. The IPMI plugin collects BMC sensor data at 20 s cadence. The documentation describes the core-temperature family as `pX_coreY_temp`, "Temperature of core n. Y in the CPU socket n. X. X=0..1, Y=0..23" [3]. *This specifies the BMC sensor-ID namespace, not the active core count*—the distinction is the entry point for this study.

III. Method

A. Recovering the disable map

For each month we enumerate the `metric=pX_coreY_temp` partitions and count samples per (node, socket, core). A core index is *present* for a socket if it emits any sample in the window. We define the disable map of socket s as $M[s,k] \in \{0,1\}^{12}$ over slices $k = \lfloor \mathrm{core}/2 \rfloor$.

The key validity argument is that **the BMC does not renumber active cores compactly**: were that so, every socket would report indices 0–15. Instead we observe arbitrary 16-element subsets of $\{0, \ldots, 23\}$, so the index is a fixed physical-position identifier.

B. External validation of the premise

Everything in this paper rests on one premise: that the set of reporting per-core sensors *is* the die's harvest map. The argument of Sec. III is internal—a renumbering BMC would yield one pattern, and we see 443. We can do better, because M100 ExaData carries two collectors that never touch the IPMI sensor family and that both know how many cores exist.

The operating system. The ganglia plugin reports `cpu_num`, the node's own logical-CPU count. POWER9 runs four SMT threads per core, so two sockets of 16 cores should present $2 \times 16 \times 4 = 128$ logical CPUs. Across the whole record—**847,297,478 samples, 990 nodes, the 29 months in which ganglia collected this metric**—the OS reports exactly 128 on **99.9853 %** of samples. Restricted to the single month 2022-08 used for the thermal work (40.9 M samples, 988 nodes) the figure is **99.997 %**. We give both because both appear below, and they are different windows rather than different measurements of the same one.

The scheduler. SLURM's configured-CPU totals across the four partitions are $110{,}208 + 12{,}672 + 2{,}560 + 256 = 125{,}696$, which is 982×128. The scheduler agrees, from a third independent path.

The one exception is itself a confirmation. Exactly one node deviates *in the month 2022-08*: node 145 reports 124 logical CPUs across 2022-08-11–12. That window contains the *only* BMC-visible reconfiguration on that node—socket p1 exchanges the slice at cores $\{22, 23\}$ for the one at $\{18, 19\}$ on 2022-08-12, while socket p0 is unchanged all month. Two collectors with nothing in common flag the same node in the same 48 hours. That is the strongest single piece of evidence we have that the sensor-presence signal tracks real hardware configuration rather than an artefact of the monitoring stack.

The premise is therefore not merely internally consistent but externally confirmed, at fleet scale, by the OS and the scheduler independently.

C. Pipeline, filters, and what each filter protects against

The recovery above is one line of logic; making it reliable at 1.67 M socket-days is mostly a question of filters, and we state them here rather than introducing them where they are first used.

Stage 1 — extraction. Each monthly archive is scanned once and only the 48 `p{0,1}_core{0..23}_temp` partitions are extracted. A **completeness gate** refuses to emit a month unless all 48 are present. This is not defensive boilerplate: a partial extraction is indistinguishable, downstream, from a socket with disabled cores, and an early version of this pipeline produced exactly that artefact for three months (Sec. III-D).

Stage 2 — daily aggregation. Samples are reduced to a count per (node, socket, core, day), giving a 15 MB table from a 400 GB scan. Every structural and longitudinal result in this paper is computed from that table; only the thermal analysis returns to the raw series.

Stage 3 — the clean-day filter. A socket-day is *clean* when exactly 16 cores report *and* each reports more than 90 % of

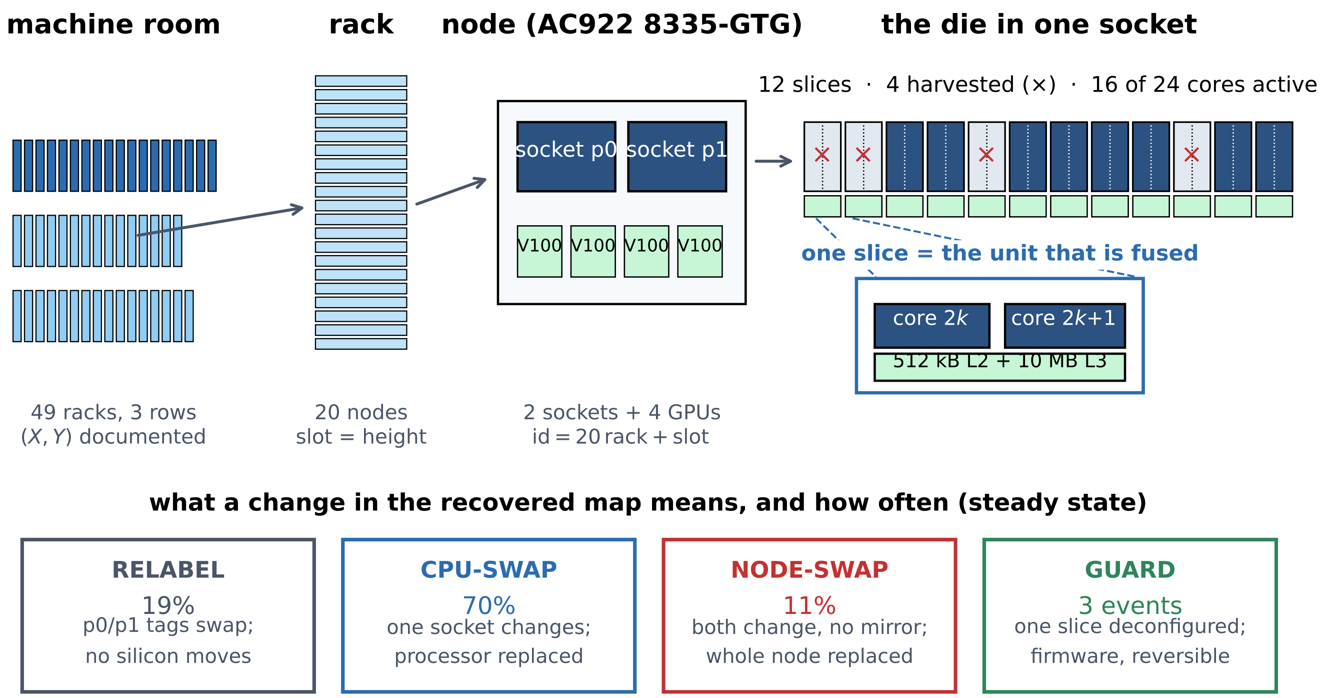

Fig. 1. The physical hierarchy this paper reasons over, and what a change in a recovered map means at each level. A socket's harvest map identifies the *die currently seated in that position*, so a change in the map is a change of die, of position labelling, or of configuration— and Sec. IV-I shows these are distinguishable from telemetry alone.

the 4,320 samples expected at 20 s cadence. This is the load-bearing choice in the paper, so its effect is stated plainly: it retains 1,517,473 of 1,671,297 socket-days (90.8 %), and Sec. IV-G shows the lot result is unchanged when the threshold is tightened to sockets with $\geq$90 % clean-day coverage. The filter exists because a partially-collected day mimics a configuration change.

Stage 4 — per-socket map. A socket is summarised by the modal harvest map over its clean days (median 775 days supporting each map). Sockets whose modal map does not have exactly four harvested slices are excluded; this removes 6 of 1,968.

Cost. One full pass is a 400 GB sequential scan, roughly two hours on a single node, dominated by archive traversal rather than computation. Re-running every result from the derived table takes minutes. The asymmetry matters for reproducibility: we publish the derived table, so the expensive stage need not be repeated.

D. Failure modes of the method

Four things could make this recovery wrong, and each has a detector.

Partial ingestion. If archive extraction silently truncates, absent metrics look like harvested cores. *Detector:* the 48-metric completeness gate. We found this the hard way — three months were once written from incomplete extractions, one of them missing all 24 sensors of socket p1, which would have read as a socket with no active cores at all.

Compact renumbering. If a BMC renumbered surviving cores $0..15$, the method would recover nothing but a constant. *Detector:* we observe arbitrary 16-element subsets of $\{0, \ldots, 23\}$ and 443 distinct patterns; a renumbering BMC would produce exactly one.

Per-metric collector faults. A collector that drops individual metric streams mimics deconfiguration. *Detectors:* odd active-core counts, which slice-granular fusing cannot produce (39 socket-days); reduced counts that are strict subsets of the socket's modal map; and, for the guard episodes specifically, the control that every other sensor on the same socket kept full cadence (Sec. IV-H).

Socket-tag swaps. If the collector exchanges the p0/p1 tags, a longitudinal study silently mixes two dies. *Detector:* the mirror test of Sec. IV-I, which identifies 19 % of steady-state map changes as relabelling rather than replacement.

E. Null model for clustering

Every socket has exactly four harvested slices, so the constraint induces negative dependence and a naive independence null is invalid. We use the curveball algorithm [8], which preserves *both* row sums (4 per socket) and column sums (per-slice marginals) while randomising association structure. We use 200 draws of 20,000 swaps each, and assert row- and column-sum preservation on every draw. Repeating the whole null at 25–400 draws moves the resulting z by about ± 1, so we quote it to the nearest integer rather than to a decimal.

F. Thermal probing of die topology

Cores on a socket share workload and inlet air, producing a dominant common mode. We remove it by subtracting the per-timestamp cross-core mean, double-centre, and compute residual correlations at the sensors' native 20 s cadence—no resampling is applied, and Sec. IV-E shows why that matters. Physically abutted cores should retain positive residual coupling.

Caveat: Common-mode removal forces each row of the residual matrix to sum to approximately zero, so negative values at large index gaps are partly mechanical. We therefore rely on contrasts *at identical index distance*, which control for this exactly.

IV. RESULTS

A. Every socket is 16 cores; granularity is the slice

Populations: Three socket populations recur and are kept distinct throughout: $\mathcal{S}_{\text{all}}$, the 1,968 sockets appearing anywhere in the record (984 nodes); $\mathcal{S}_{\text{map}}$, the 1,962 with at least one clean day and hence a well-defined harvest map; and $\mathcal{S}_{2004}$, the 1,960 present in the single month 2020-04. All structural results below are over $\mathcal{S}_{\text{map}}$ and the full 31-month record. One further distinction matters only for the rack-level analyses: node identities run to 983, four beyond the 980 that fill the 49 documented cabinets, and those four have no documented (X, Y) floor position. The 2 sockets of theirs that carry a map are excluded wherever a result is indexed by rack, leaving 1,960; being a four-node group at the end of the id range, they would otherwise form a degenerate segment that captures the argmax of any unguarded changepoint scan (Sec. IV-G).

TABLE I
DISABLE-MAP STRUCTURE OVER THE FULL 31-MONTH RECORD.

Sockets with a well-defined harvest map	1,962
Median clean days supporting each map	775
Socket-days reporting exactly 16 cores	**99.9847%**
Sockets ever reporting 24 cores	0
Harvested cores form complete $(2k, 2k+1)$ pairs	**1,962 (100%)**

The perfect pair structure is the central mechanical finding: it matches the POWER9 slice definition exactly and rules out per-core fusing.

B. The map is per-die, not per-SKU

443 distinct patterns appear among the $\binom{12}{4} = 495$ possible (Fig. 8). The most common (slices 0–3) covers only 5.5% of sockets. Within a node the two sockets match in just 1.2 % of cases (12 of 981 nodes), with mean overlap 1.641 slices against a shuffled null of 1.468 ± 0.026 ($z = +6.7$)—nearly independent, with a small excess attributable to same-lot pairing.

C. Marginals are strongly non-uniform

Slice 0 is harvested on 61.2% of sockets against a uniform expectation of 33.3% ($\chi^2 = 807.2$, df $= 11$, $p \ll 0.001$); Fig. 3 splits this by lot. These marginals are remarkably stable in time: across the 31 months the largest standard deviation of any slice's monthly rate is **0.25 percentage points**, and slice 0 ranges only 60.9–61.8% (Fig. 12). The single-month snapshot used in earlier drafts was therefore representative, and every estimate here is a 2.5-year-stable measurement.

D. Harvested slices cluster in index space

Under the curveball null the mean index gap between harvested slices is 4.242 versus 4.582 ± 0.017 ($z \approx -20$; Monte-Carlo noise ± 1, see Sec. III-E). Applying a Bonferroni correction across all $\binom{12}{2} = 66$ pair tests ($|z| > 3.83$) leaves 16–18 pairs significant (Fig. 4).

That range is not vagueness but the honest precision of the statistic. The same Monte-Carlo noise that forces $z \approx -20$ above also moves individual pairs across the threshold: repeating the whole null under five seeds gives 16–18 significant pairs, of which 8–9 are positive excesses and 8–10 deficits. **What does not move is the structure.** In every run, *exactly eight* positive-excess pairs have $|\Delta k| \leq 2$—the six adjacent pairs 0&1, 2&3, 4&5, 6&7, 8&9, 10&11 plus 1&2 and 0&2, with 0&1 always the strongest at $z \approx +15$ to $+17$. Whether a ninth, more distant pair also clears the threshold varies by seed. Deficits are overwhelmingly distant—$|\Delta k| \geq 4$ in almost every case—though a single near pair (2&4) crosses the threshold in some runs, and we report that rather than describing the deficits as uniformly distant.

Because Sec. IV-F shows index adjacency *is* physical adjacency, this is spatial clustering, not merely index-space clustering.

How big, not just how significant: A z is a signal-to-noise ratio and grows with $\sqrt{n}$, so the values above report sample size as much as effect. Fig. 4 therefore plots the effect itself, $\log_2$ of observed over expected co-harvest, against slice distance. The excess is real but moderate: averaged over pairs at $|\Delta k| \leq 2$ it is $+0.165$ ($1.12\times$ the null), against -0.089 ($0.94\times$) for more distant pairs, with the strongest individual pairs reaching 1.7–1.8×. Clustering of this size is easy to detect across 1,962 sockets and easy to overstate from a z alone, so both are given.

E. Slice siblings are physically abutted

We probe **every** socket with both halves fully configured—**1,948 sockets across 974 nodes**—over the full 31-day month 2022-08 (Fig. 5). At *identical* index distance of 1, within-slice siblings are more strongly coupled than cross-slice neighbours: $r = +0.372$ (95 % CI $[0.369, 0.376]$, $n =$ 15,584 pairs) against $+0.312$ ($[0.306, 0.317]$, $n =$ 10,122), a **contrast of** $+0.061$ with permutation $p < 10^{-5}$ and non-overlapping intervals. It holds on **69 % of individual sockets** taken one at a time, so the pooled figure is not carried by a minority.

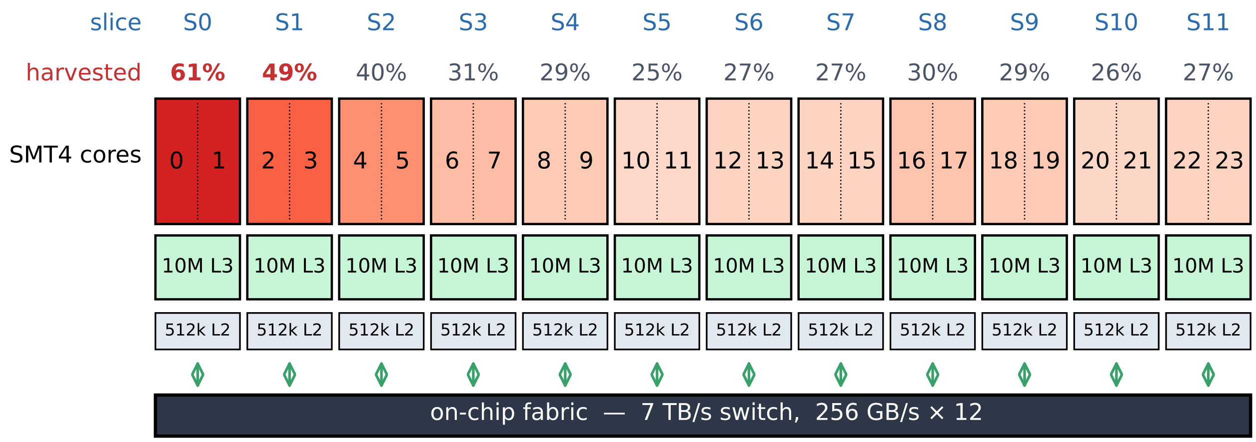

Fig. 2. POWER9 slice organisation (after [5]) annotated with the measured per-slice harvest rate across the 1,962 Marconi100 sockets with a well-defined map. Each slice holds two SMT4 cores sharing 512 kB L2 and 10 MB L3 on a common fabric port. Harvesting always removes whole slices. **The horizontal ordering is the IPMI sensor index, which Sec. IV-F shows is the physical die ordering, recovered independently from thermal coupling.**

A census, not a sample: An earlier version used 120 nodes, bounded by memory rather than by design, and reported $+0.377$ vs $+0.308$ on 240 sockets. Subsampling in *time* rather than in nodes—keeping every fifth 20 s sample, which lowers the sample count without smoothing and so, unlike a coarser alignment grid, does not inflate correlations—lets the whole fleet be processed in one pass. The census moves the contrast by 0.008, and the floorplan fit from -0.787 on the 180-socket subsample of Table III to -0.791 fleet-wide. (A third value, -0.754, appears in Sec. IV-F; it belongs to the separate 240-socket subsample used there. The three are different populations, not successive corrections of one.) Every thermal number reported here is a fleet-wide measurement. The two cores of a slice are measurably more coupled than equally-numbered cores across a slice boundary, confirming physical abutment and the slice as a physical unit.

The effect size depends on temporal resolution; the conclusion does not: Cores are sampled every 20 s and their timestamps coincide across a socket, so no resampling is required—an inner join at native resolution retains 133,907 of 133,920 possible rows (99.99 %). Earlier versions of this analysis nonetheless aligned onto a one-minute grid, a step introduced to work around an unrelated bug and never revisited. It is not neutral (Table III): coarser averaging suppresses high-frequency noise and inflates every correlation, within-slice more than cross-slice, so within this subsample the contrast grows from $+0.063$ at 20 s to $+0.140$ at 15 min. **We therefore quote the native-resolution figure, which is the most conservative.** What does *not* move is the ordering—within-slice exceeds cross-slice at every grid—and, critically, the floorplan fit of Sec. IV-F, which sits at -0.787, -0.773, -0.766, -0.767 across the four grids. The paper's load-bearing thermal claim is independent of this choice.

Abutment or shared cache?: The two cores of a slice are physically adjacent *and* share an L2/L3 block, so shared *activity* is an alternative explanation for the excess that would need no thermal path. Core-to-core thermal time constants on a die are far shorter than a 20 s sample, so both mechanisms peak at lag zero; what distinguishes them is the *shape* of the lagged profile. Instantaneous activity sharing predicts a sharp lag-0 peak decaying with the workload's own autocorrelation. We measure the opposite (Table II): over all 1,948 sockets the excess is *smallest* at lag 0 and grows monotonically to $1.60\times$ that value at ±10 samples (200 s), while the cores' own autocorrelation falls to 0.51 over the same span. A broad, lag-persistent excess is the signature of a slowly varying shared physical environment. (Both curves spike at lag 0 relative to their neighbours, consistent with a shared same-sample measurement artefact—so the excess is least trustworthy at exactly the lag an activity explanation would need it to be largest.)

TABLE II
LAGGED CROSS-CORRELATION.

lag (20 s)	within	cross	excess	core autocorr.
0	0.3723	0.3118	0.0606	1.000
±1	0.2930	0.2102	0.0828	0.613
±2	0.2960	0.2068	0.0892	0.587
±5	0.2948	0.1995	0.0953	0.542
±10	0.2902	0.1932	0.0970	0.505

All 1,948 sockets, common mode removed. The within-slice excess *grows* with lag; instantaneous activity sharing predicts the opposite.

A correction to an earlier estimate: An initial version of this analysis used 40 sockets in 2020-04, a month with only three reporting days, and reported a contrast of $+0.51$ vs $+0.29$. That overstated the effect roughly threefold. The direction, the significance and the replication across sockets survive; the magnitude does not. We report the full-month estimate.

TABLE III
THERMAL RESULTS AGAINST ALIGNMENT GRID.

Grid	rows/socket	within	cross	1×12 fit
20 s (native)	**133,907**	**+0.377**	**+0.314**	−0.787
1 min	44,640	+0.411	+0.329	−0.773
5 min	8,928	+0.466	+0.351	−0.766
15 min	2,976	+0.508	+0.368	−0.767

This table is a 180-socket subsample, held fixed across the four rows so that only the grid varies. Its 20 s row therefore differs slightly from the 1,948-socket census reported in the text (+0.372/+0.312, fit −0.791). The point of the table is the trend down the rows, not the absolute values.

F. The sensor index recovers the physical slice ordering

Is the sensor index merely a label, or does it track physical position? We answer it by scoring explicit candidate floorplans: for each, correlate the predicted physical distance between two slices against their measured thermal coupling (Fig. 6). Coupling should fall with distance, so a better floorplan gives a more negative correlation.

Over all 1,948 sockets the identity 1×12 linear arrangement wins decisively, at -0.791 against a random-ordering null of $+0.001 \pm 0.121$ ($z = -6.55$, $p < 10^{-4}$), and beats every two-dimensional alternative tried (6×2: -0.742; 4×3: -0.578; 3×4: -0.481; 2×6: -0.238). More stringently, an unconstrained hill-climbing search over one-dimensional orderings—free to return any of the $12!/2$ arrangements—converges to $[10, 11, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0]$ at -0.793, barely better than the identity's -0.791. That is the sensor ordering *reversed*; since distance is invariant under reflection it is the same physical arrangement, differing by one adjacent transposition. Rank correlation with the identity is $|\rho| = 0.993$.

Every number in this paragraph comes from the 1,948-socket census, scored on one matrix. We state that explicitly because the decisive comparison here is a *margin*—hill-climb against identity—and a margin computed across two populations would mean nothing. On the 240-socket subsample the same two values are -0.759 and -0.754: a different population, the same negligible gap.

This result cannot be cache sharing: The slice-level coupling above is computed from **cross-slice pairs only**, and two cores in different slices share no L2 and no L3. Measured over those pairs alone, coupling decays monotonically with slice distance—from $+0.215$ at adjacent slices to -0.353 at distance 10, giving corr(distance, coupling) $= -0.929$. Whatever explains the within-slice excess, it cannot explain this: the pairs that share cache are excluded from the floorplan estimate by construction.

The thermal data therefore recovers the sensor-index ordering independently, up to reflection. This licenses reading Sec. IV-D spatially: slices adjacent in index are adjacent on the die, so the co-harvest excess is genuine *spatial* clustering of the kind the yield literature describes [9], [10].

A correction to an earlier estimate: On the three-day 2020-04 sample this test was inconclusive—the linear ordering scored only -0.283 ($p = 0.023$) and was beaten by a 3×4 grid—and we previously declined to claim a physical ordering. The full-month, both-socket measurement reverses that conclusion. We report both so the reader can judge how much the earlier sample size mattered.

G. A procurement-lot boundary at rack 22

What a rack is here: Marconi100's nodes occupy 49 physical rack cabinets of 20 nodes each, with $\mathrm{id} = 20\,\mathrm{rack} + \mathrm{slot}$, and the dataset documents every rack's (X, Y) position on the machine-room floor in three rows [3]. The rack index is a real physical coordinate, not a label—which is why it is informative, and why it must be handled carefully: it conflates *procurement batch* (hardware arrives and is racked in deliveries, in id order) with *room position* (cooling, airflow, height in cabinet). We separate the two below.

A changepoint scan maximises Welch t at rack 22, $t = 26.6$ (Fig. 7). At node granularity the boundary is sharper still: $t = 26.8$ between nodes 438 and 439, with the 40 sockets immediately before at 95.0 % slice-0 harvested and the 40 immediately after at 42.5 %. Node 439 sits in the topmost slot of rack 21 yet behaves like the later population (0 of its 2 sockets have slice 0 harvested, against 2 of 2 for each of nodes 430–438), so the delivery boundary is rack-aligned to within a single node.

TABLE IV
TWO POPULATIONS EITHER SIDE OF THE LOT BOUNDARY, AT THE TWO GRANULARITIES THE BOUNDARY CAN BE DRAWN AT.

Cut	Population	Sockets	P(slice 0)	Patterns
rack	Lot A (racks 0–21)	880	88.2%	230
	Lot B (racks 22–48)	1,082	39.3%	416
node	Lot A (nodes 0–438)	878	88.4%	229
	Lot B (nodes 439–979)	1,084	39.2%	416

The two cuts differ by exactly one node. Racks are 20 nodes wide, so the rack cut places node 439—the topmost slot of rack 21—in Lot A, whereas the node-level changepoint assigns it to Lot B, which is where its harvest map says it belongs (0 of its 2 sockets have slice 0 harvested, against 2 of 2 for each of nodes 430–438). Nothing downstream turns on the choice; we use the *rack* cut throughout, because the lot-level model of Sec. IV-J resamples racks, and we state it here so the two socket counts are not mistaken for a discrepancy.

Lot A marginals fall steeply with slice index; Lot B is nearly flat. Permutation tests confirm between-rack heterogeneity for slices 0, 1, 2, 5 ($p < 0.01$), and same-rack sockets share more harvested slices than chance (1.565 vs 1.452 ± 0.008, $z = +13.9$).

The boundary, tested rather than maximised: A t of 26.6 obtained by scanning every split is a *maximum* over dependent tests, and cannot be read as a significance level. We therefore calibrate it. Permuting whole racks—which preserves rack sizes and the within-rack dependence documented above—the null distribution of $\max|t|$ has mean 7.08 ± 2.41 and never exceeds 17.95 in 4,000 draws, against the observed 26.6 (Fig. 9): calibrated $p < 2.5 \times 10^{-4}$. Sockets resampled within rack (racks cannot be resampled with replacement, since a changepoint is defined on their order) put the breakpoint at rack 22 in **100 % of 2,000 resamples**, a 95 % CI of $[22, 22]$. Leaving out each rack in turn recovers rack 22 in **48 of 49** fits. The scan is restricted to boundaries leaving at least 100 sockets per side; without that guard the argmax is captured by a four-node partial rack at the end of the id range, which we exclude and report (Sec. IV-A).

Two deliveries, not more—chosen out of sample: Rather than a stopping rule on $|t|$, we segment the rack-ordered fleet into k populations by dynamic programming over a per-segment twelve-slice Bernoulli likelihood, and choose k by *rack-held-out* likelihood, since in-sample fit always rewards more segments. Held-out log-likelihood per socket is -7.360 at $k = 1$, -7.133 **at** $k = 2$, then -7.139, -7.157 and -7.149 at $k = 3, 4, 5$. Two populations is the out-of-sample optimum, and the $k = 2$ cut is rack 22. The fleet is two deliveries—not a gradient, and not a sequence of many small batches.

The lot is predictable, not merely detectable: The operationally meaningful question is whether a harvest map assigns a rack the model has never seen to the right delivery. Holding out each rack in turn and classifying its sockets by likelihood ratio against the two lot models fitted on the other 48, **72.9 % of individual sockets** are assigned correctly against a 55.1 % majority-class baseline, and **47 of 49 held-out racks (95.9 %)** are assigned correctly by majority vote. Provenance is recoverable prospectively, which a large in-sample t alone would not establish.

Procurement, not room position: Because rack index conflates batch with geometry, we test the geometric alternative directly. Raw correlations with slice-0 status are substantial for the machine-room row ($r = -0.414$ for Y) and visible for aisle position ($r = +0.124$), exactly as expected when both track rack index. **Once the lot is accounted for, they vanish**: partial correlations are $+0.017$ for Y ($p = 0.44$) and $+0.014$ for X ($p = 0.53$). Only the slot—height within the cabinet—retains a residual -0.054 ($p = 0.017$, 0.3 % of variance), which is marginal under a three-test correction and would in any case be consistent with racks being filled in delivery order.

This is the empirical form of an argument that also holds *a priori*: fusing happens at manufacture, months before a part is installed, so no property of the machine room can influence a harvest map. The only admissible causal direction runs from procurement batch to rack position.

The boundary is not an artefact of the coverage filter: Because every structural result is computed on *clean* socket-days, a lot-correlated dropout could in principle manufacture this contrast. It does not. Clean-day rates are near-identical between lots (0.909 vs 0.902), coverage is uncorrelated with the harvest pattern (corr with slice-0 status = $+0.004$, $p = 0.88$), and restricting to the 1,847 sockets with $\geq$90 % clean-day coverage leaves the contrast unchanged at $+49.0$ percentage points versus $+48.9$ over all sockets.

H. Configuration lifetime

POWER9 exposes mechanisms that *could* alter the active set after manufacture: predictive Dynamic Processor Deallocation, persistent GARD records read by hostboot at IPL and clearable with `opal-gard`, and the field core override used for licensing [11], [12]. Operating-system core offlining is invisible to the BMC and cannot affect sensor presence. Whether harvest maps are static in practice is therefore an empirical question, which we answer by aggregating every core sample to (node, socket, day) counts across the dataset. All numbers in this section are over the complete 31-month record (2020-03 to 2022-09).

The count is very nearly invariant: Across all 31 months—1,671,297 socket-days over 858 reporting days and 984 nodes—**99.9847% report exactly 16 active cores**. Of the 256 exceptions, 39 have an odd count, which *cannot* arise from slice-granular fusing and therefore proves the collector can drop individual cores; the counts above 16 are unions spanning a same-day reconfiguration. Only **nine** socket-days are fully-sampled with an even count below 16, and these are the subject of the guard analysis below.

Configurations are near-static: Restricting to *clean* socket-days (16 cores, each at >90% of the 4,320 samples expected at 20 s cadence) leaves 1,517,473 socket-days, a median of 775 observed days per socket. **89.5% of sockets never change configuration** (95 % CI $[88.1, 90.8]$) (1,756 of 1,962); 183 have two distinct sets, 19 have three, three have four and one has five.

Three core-guard episodes: The nine anomalous socket-days resolve into **three sustained episodes on three distinct sockets**, each with the exact signature of a field deconfiguration (Table V). In every case the socket loses *exactly one slice*—a complete $(2k, 2k+1)$ pair—runs on 14 cores for 2–4 consecutive fully-sampled days, and in at least one case the slice is subsequently restored, consistent with an operator clearing the GARD record [12].

Rates need care here, in two respects. First, **one of the three episodes (347 p1, 2020-03-13–16) falls inside the commissioning window** that Sec. IV-H excludes; the other two are steady-state. We therefore quote **two steady-state episodes**, and report the third separately rather than folding an excluded-period event into a steady-state rate. Second, an episode is counted only on fully-sampled days, so one falling entirely inside a coverage gap would be missed; the appropriate denominator is *observed* exposure, which over the steady-state window is 3,991 socket-years. That gives **5.0 episodes per 10,000 socket-years**, exact Poisson 95 % CI $[0.6, 18.1]$—an interval spanning a factor of thirty. **This is an order-of-magnitude statement, not a precise rate**, and we state it as one. Durations are likewise lower bounds.

Independently confirmed by the operating system. The deviations in the OS core count are not noise, and they are not uniform. Of 990 nodes, exactly **three** ever report **120** logical CPUs—which is $128 - 8$, precisely one slice of two cores at four threads. Those three nodes are **347, 411 and 946**: the same three sockets, and only those, that the BMC analysis identifies as core-guard episodes. Under a null in which the OS flagged three arbitrary nodes, the chance of matching our three is $1/\binom{990}{3} = 6.2 \times 10^{-9}$.

The dates agree too, and agree in the direction the method predicts. For node 411 the OS window is 2021-11-12–16 against a BMC episode of 11-13–15; for node 946, 2020-08-

30–09-02 against 08-31–09-01. In both cases the OS window *brackets* the BMC one, exactly as it should, since a BMC episode is counted only on fully-sampled days and its duration is therefore a lower bound. Node 347 additionally shows an OS deviation in 2020-10 that our strict criterion did not flag, which if anything means we undercount.

This is the strongest evidence in the paper. Three events recovered from thermal-sensor presence alone are confirmed, node-for-node, by the operating system's own CPU count—a quantity produced by a different collector, from a different layer of the stack, that never touches a BMC sensor.

And the OS sees what the BMC cannot. Six further nodes report **124** logical CPUs ($128-4$, one SMT4 *core*). That is not slice-granular and so cannot be a fusing event; on those same days the BMC still reports its usual 16 cores. This is operating-system core offlining, invisible to the BMC—precisely the layer separation asserted in Sec. IV-H, here observed rather than argued.

Not an artefact of a tight filter. The criterion above is strict, so we re-ran the search against the *adjacent observed day* rather than the modal map, requiring only that surviving slices exceed 50 % sampling rather than 90 %. It returns **the same three episodes and no others**, two of which show the slice returning within 30 days. Relaxing the filter does not multiply the count, so three is a property of the fleet and not of the threshold. (A first attempt compared each day against the socket's *global* modal map instead; that conflates guard events with permanent reconfigurations—a socket that changes configuration mismatches its modal map for hundreds of days—and inflated the count to 30. We report the failed variant because the same mistake would inflate anyone else's count the same way.)

Ruling out a collector fault. The competing explanation is that the collector dropped exactly those two metric streams. We test it directly: for each episode we ask whether the *other* sensors on the same node—in particular on the same socket—kept reporting. They did. In all three episodes the two slice sensors read zero for the entire window while **all ten control metrics, including `pX_power` and `pX_vdd_temp` on the same socket, sustained a full 4,320 samples per day** (Fig. 13). A collection fault that silences precisely the two cores of one slice for several consecutive days, while leaving every other sensor on the same socket at full cadence, is not a plausible account.

And they sit across reboots, as the mechanism requires. A GARD record is applied by hostboot at IPL, so a deconfiguration—and its later clearing—should each land on a boot. Using the independent `boottime` signal of Sec. IV-H, both steady-state episodes are bracketed by observed reboots: node 946's episode of 2020-08-31–09-01 falls between boots on 08-30 and 09-03, and node 411's of 2021-11-13–15 between boots on 11-12 and 11-15/16. The third episode (node 347, 2020-03) cannot be checked at all, because ganglia `boottime` collection begins on 2020-05-05—an absence of coverage, not an absence of a reboot, and one more reason that episode is reported separately from the steady-state rate.

We therefore attribute these episodes to hardware deconfiguration; to our knowledge this is the first direct observation of POWER9 field core guarding in a deployed system.

TABLE V
CORE-GUARD EPISODES.

Socket	Dates	Days	Slice lost	Restored
347 p1	2020-03-13–16	4	4 (cores 8,9)	yes
411 p1	2021-11-13–15	3	6 (cores 12,13)	yes†
946 p0	2020-08-31–09-01	2	8 (cores 16,17)	not observed

Baselines are the adjacent observed day, not the socket's global modal set, since two of these sockets also change configuration elsewhere in their life. †Restored via a day that also spans a reconfiguration.

Changes never occur during operation: Of **235** transitions across the full record, **zero** occur between consecutive reporting days: every one coincides with a gap in reporting (median 2 d), against a background in which only 4.2% of consecutive reporting days are more than one day apart.

A reporting gap is not a reboot, so we measure the reboot: That argument infers a reboot from a *gap*, and collectors stop for many reasons. The inference is unnecessary: the ganglia plugin publishes `boottime`, the epoch at which a node last booted, so a reboot is directly observable as a change in that value. Extracting it for all 31 months gives 668,524 node-days across 990 nodes, of which 1.74 % carry a reboot.

Of the **132 steady-state transitions with boot coverage, 126 (95.5 %) coincide with an observed reboot**, against **1.74 %** of 1,194,581 non-transition intervals. The comparison must be matched on interval length, since a longer gap has more opportunity to contain a boot; it survives that (Table VI), and a permutation test that shuffles the reboot flag *within* gap strata gives an observed difference of $+93.7$ percentage points against a null of $+41.3 \pm 3.5$ ($p < 2.5 \times 10^{-4}$).

TABLE VI
MAP TRANSITIONS AGAINST AN INDEPENDENT BOOT SIGNAL, MATCHED ON INTERVAL LENGTH.

Interval	Transitions with reboot	Controls with reboot
2 d	93.0 % ($n = 71$)	35.8 % ($n =$ 16,055)
3–4 d	100 % ($n = 20$)	16.7 % ($n =$ 21,953)
5–8 d	100 % ($n = 22$)	12.9 % ($n =$ 1,864)
≥9 d	94.7 % ($n = 19$)	66.3 % ($n =$ 5,912)

No transition has a 1-day interval, which is the same fact as "zero occur between consecutive reporting days".

The claim is therefore no longer that changes coincide with missing data, but that they coincide with *boots* [OI-14], measured by a collector that never touches the IPMI sensor family. The remaining 4.5 % are cases where the node rebooted outside the window our daily granularity can resolve, not counterexamples we can attribute to in-operation change.

The only two apparent counterexamples confirm the rule once examined at full timestamp resolution. Nodes 315 (p1) and 342 (p0) appear to change within a single day, and produce the dataset's only 20- and 22-core socket-days. In fact both cut over on 2021-03-02 across a *sub-day reporting* *gap*: on node 315 the outgoing cores $\{4, 5, 20, 21\}$ report until 08:42 and the incoming cores $\{8, 9, 10, 11\}$ resume at 09:10, a 28-minute outage, with exactly 16 cores before and 16 after; the 20/22-core counts are unions across the cutover. Node 342 cuts over in the same window, indicating a coordinated maintenance reboot. Day-granularity aggregation is therefore *conservative* here: it cannot see sub-day downtime, so our transition-gap statistics understate how consistently changes coincide with reboots.

Staticity as a prediction, not a description: The figures above describe the record in-sample. We also test them out-of-sample: derive each socket's map from the *first* half of the record (2020-03 to 2021-06) and use it to predict every clean socket-day in the *second* (2021-07 to 2022-09). Over **839,067 held-out socket-days the prediction is correct 96.77 % of the time** (95 % CI $[96.73, 96.81]$), and the errors are not diffuse: they fall on **89 of 1,962 sockets** (4.5 %), which are the sockets that genuinely reconfigure between the two halves. A map measured once therefore forecasts a socket more than a year later.

Changes concentrate at commissioning: The 235 transitions involve 132 nodes on 81 distinct dates, but are heavily clustered: 2020-03-20 alone accounts for 63 and the top three dates for 41 %. Such synchronised, fleet-wide changes are characteristic of a newly commissioned machine—hardware is still being reconfigured and the monitoring pipeline is not yet settled—rather than of steady-state operation. The next paragraph shows this directly and excludes the period.

Commissioning period, identified and excluded: Two independent signals locate the end of commissioning without assuming a date. Telemetry coverage—the fraction of socket-days that are clean—is 0.82, 0.00 and 0.40 in 2020-03, -04 and -05, then rises to 0.72–1.00 for every one of the following 28 months. The transition rate falls from 91 in 2020-03 to a median of 4 per month thereafter. We therefore treat 2020-03 to 2020-05 as commissioning and re-run every result on 2020-06 onward:

- **The harvest maps are unchanged**: for all 1,960 sockets with a map in both windows, the recovered map is *byte-identical* (100.0 %). Marginals agree to 0.1 percentage points and the pattern count is 443 either way.
- **The lot boundary is unchanged**: Lot A 88.2 %, Lot B 39.2 %, Welch $t = 26.6$.
- **Zero transitions during operation still holds**: of the 136 steady-state transitions, *none* occurs between consecutive reporting days.
- **The fleet-wide clustering disappears with the commissioning window**: the largest single-date cluster in steady state is 11 events (8 % of steady-state transitions), against 63 (27 %) on 2020-03-20.

Every structural and longitudinal claim in this paper therefore rests on the post-commissioning period, and none of them depends on the anomalous early months.

Some apparent changes are socket relabelling: When node n's socket p0 loses exactly the slices p1 gains and vice versa, the dies are unchanged and only the p0/p1 labels have swapped. This accounts for **19%** of steady-state changes (20 of 105; 95 % CI $[12, 28]$) and 20% of the 230 transitions across the whole record where both sockets are observable.

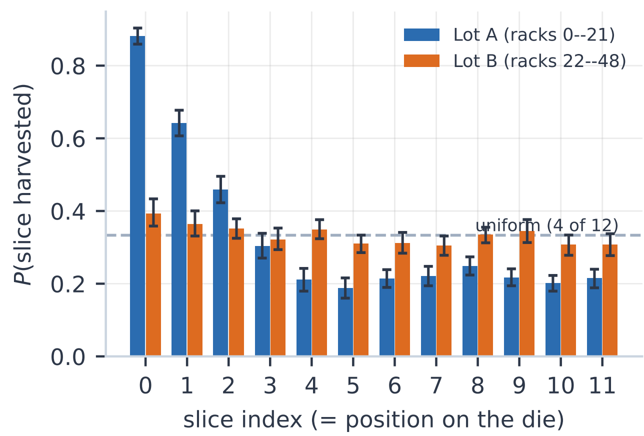

Fig. 3. Per-slice harvest probability by procurement lot, with rack-clustered bootstrap 95 % intervals (2,000 resamples of the 49 cabinets). Lot A falls steeply with slice index; Lot B is nearly flat.

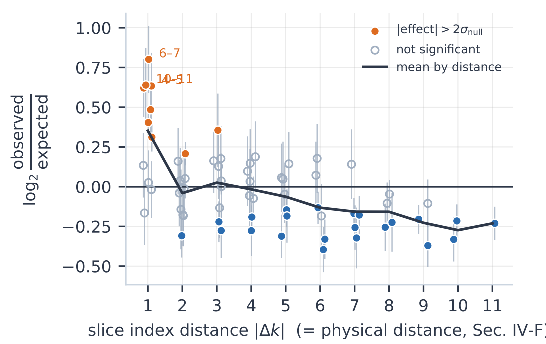

Fig. 4. Co-harvest effect size against the curveball null, as $\log_2$(observed/expected), for all 66 slice pairs against their distance on the die. The excess is confined to near-adjacent pairs and reverses at distance; error bars are 2σ of the null. Plotting the ratio rather than a z keeps sample size out of the vertical axis.

Caveat: An apparent change can arise from (a) genuine hardware reconfiguration, (b) socket relabelling, or (c) a change in the BMC sensor-ID mapping, e.g. after a firmware update. Telemetry alone cannot separate these. The tight clustering of 86 transitions into three days of March 2020 is consistent with (c)—but that window is precisely the commissioning period excluded above, and every result is reported on the steady-state period where no such clustering exists. The ambiguity is therefore bounded rather than load-bearing.

I. A taxonomy of hardware changes

A harvest map identifies the die in a socket, so a change in the map is informative about what physically happened—but only if the possible causes can be told apart. They can (Fig. 1, bottom). If p0 loses exactly the slices p1 gains and

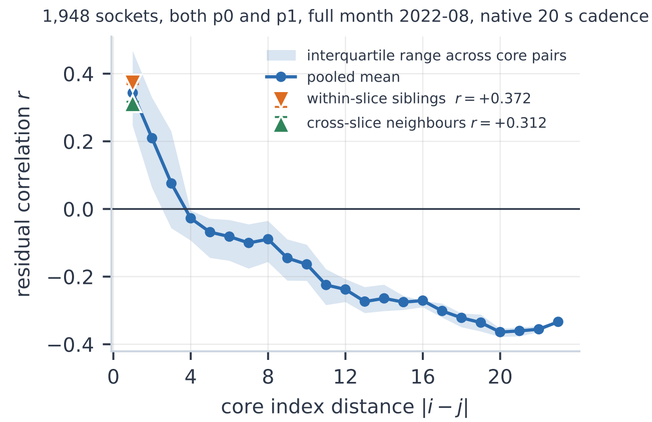

Fig. 5. Residual thermal correlation versus core index distance. The within- versus cross-slice contrast at distance 1 controls exactly for index gap.

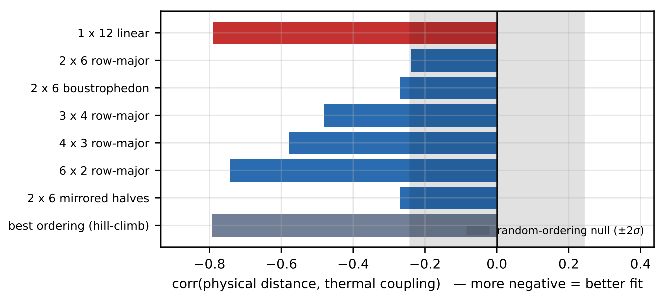

Fig. 6. Candidate floorplans scored against measured thermal coupling (more negative = better fit). The linear index ordering (red) wins decisively and is matched only by an unconstrained search that converges to the same arrangement reversed.

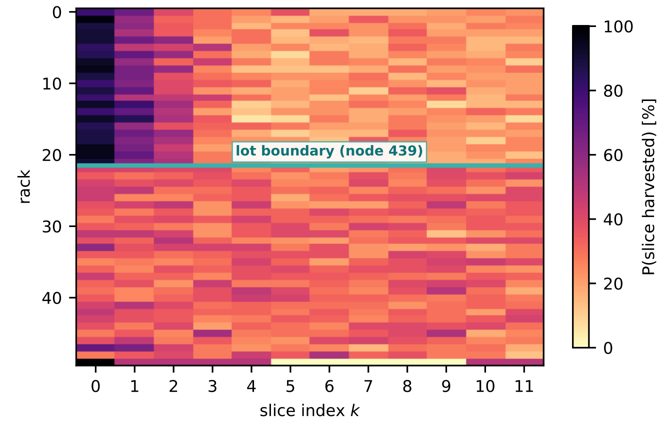

Fig. 7. Harvest probability by rack (rows: the 49 physical cabinets, 20 nodes each, in installation-id order) and slice (columns: position on the die). The procurement boundary is visible as an abrupt change across the whole die at once—the signature of a different population of parts, not of a room effect.

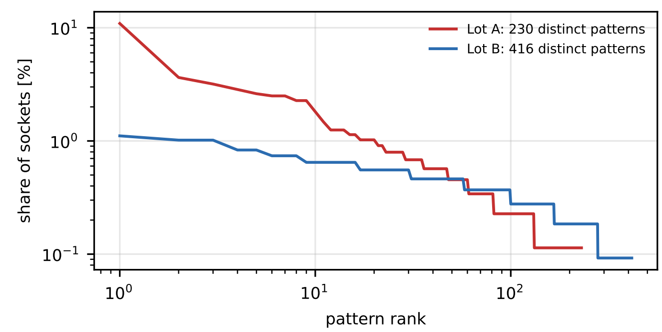

Fig. 8. Rank–frequency of disable patterns. Lot B is markedly more diverse.

vice versa, the same two dies are present and only the tags moved (RELABEL). If exactly one socket's map changes while the other is observed and unchanged across the same interval, one processor was replaced (CPU-SWAP). If both change and it is not a mirror, the node or both processors were replaced (NODE-SWAP). And a count dropping below 16 by exactly one slice, later restored, is a firmware deconfiguration with no hardware movement at all (GUARD, Sec. IV-H).

TABLE VII
HARDWARE-CHANGE EVENTS, CLASSIFIED AT NODE LEVEL.

Kind	Commiss.	Steady
RELABEL	3	20
CPU-SWAP	11	68
CPU-SWAP?	2	6
NODE-SWAP	40	11
Total	56	105

RELABEL: no silicon moved. CPU-SWAP: one processor replaced.
CPU-SWAP?: as above, partner socket unobserved across the interval.
NODE-SWAP: node or both processors replaced.

The split validates the taxonomy on its own terms. During commissioning NODE-SWAP dominates (40 of 56, 71 %)—whole machines are being installed and reconfigured. In steady state the picture inverts: CPU-SWAP accounts for 70 % and NODE-SWAP for 11 %, which is what a maintenance regime that replaces individual failed processors looks like. Steady state contains 105 events on 87 distinct nodes over 28 months, about 3.8 per month.

Two consequences are worth stating. First, **19 % of steady-state map changes move no silicon**; a study that read every change as a replacement would overcount by that much. Second, counting one die per CPU-SWAP and two per NODE-SWAP gives **96 processor replacements**, recovered without any asset database or maintenance log. These are steady-state events, so the denominator must be the steady-state exposure and not the whole record: 1,960 sockets over the 850-day post-commissioning window, or 4,561 socket-years. The **annualised socket replacement rate is 2.1 %** (exact Poisson 95 % CI $[1.7, 2.6]$). Calendar exposure is the right denominator for this event, because a replacement is recovered by comparing maps *across* a reporting gap—an unobserved day is still a day at risk. Counting only observed clean socket-days instead would raise the estimate to 2.4 % $[2.0, 2.9]$, so the figure we report is the conservative one. Replacements are mildly clustered by rack (37 of 49 racks touched; concentration

$z = +2.55$), consistent with correlated failures or with servicing being batched by cabinet.

J. A clustering parameter, fitted per lot

Section IV-D establishes clustering against a null; it does not *quantify* it, and the yield models we invoke [9], [10] are parameterised by a clustering strength. Those models describe defect *counts* per die, which we cannot observe: the SKU fixes the count at exactly four harvested slices. What is observable is the conditional spatial arrangement given four, so we fit the natural conditional analogue—a pairwise-interaction model on the twelve slice positions,

$$P(S) \propto \exp\Big(\textstyle\sum_{k\in S}\beta_k + \gamma\,\mathrm{adj}(S)\Big),$$

where $\mathrm{adj}(S)$ counts index-adjacent pairs in the harvested set S. The β_k absorb per-slice propensity (the slice-0 asymmetry) and γ measures clustering *beyond* what those propensities explain. The normalising constant is an exact sum over all $\binom{12}{4} = 495$ patterns, so likelihood and gradient are exact.

TABLE VIII
FITTED CLUSTERING PARAMETER BY LOT.

Population	n	γ	LR χ^2_1 vs γ=0
Lot A (racks 0–21)	880	$+0.743 \pm 0.046$	174.2
Lot B (racks 22–48)	1,082	$+0.434 \pm 0.055$	106.4
Whole fleet	1,962	$+0.647 \pm 0.046$	394.5

Uncertainties are rack-clustered bootstrap standard errors (400 resamples of the 49 racks), not i.i.d. values, because sockets are not independent—see the text.

Three things follow. First, **clustering is real in both lots** (χ^2_1 of 174 and 106, $p \ll 0.001$): mean adjacent-pair counts are 1.463 and 1.218 observed against 1.203 and 0.993 under the no-clustering model. Second, **the lots differ significantly in clustering strength**, not only in their marginals. Here the uncertainty needs care: an i.i.d. likelihood over sockets would give $z = +5.22$, but sockets are *not* independent—each node contributes two, and Sec. IV-G shows same-rack sockets share more harvested slices than chance. Resampling *racks* rather than sockets inflates the standard errors by $1.05\times$ to $1.58\times$ and moves the difference to $\gamma_A - \gamma_B = +0.309$, $z = +4.34$, 95 % CI $[+0.183, +0.456]$. The claim survives the correct uncertainty; we report the clustered figure. Third, and most usefully, the two effects *separate*: Lot A's $\beta_0 = +2.51$ dwarfs every other propensity while Lot B's β vector is nearly flat (range $+0.34$ to -0.13), yet both carry a substantial positive γ. The slice-0 asymmetry and the spatial clustering are therefore distinct phenomena rather than two views of one.

The model is useful and incomplete, and we show both: A likelihood-ratio test tells us $\gamma \neq 0$; it does not tell us the model is adequate. We therefore simulate from the fitted model and compare it to the data on quantities it was *not* fitted to—the model is fitted to the per-slice counts and the adjacency total, so the full pattern distribution, the number of distinct patterns, and the largest gap between harvested slices are all out-of-statistic checks.

It passes some and fails others. The count of distinct patterns is reproduced (443 observed against 436.6 ± 6.2 replicated, $p_{\text{ppc}} = 0.17$). But the pattern-frequency discrepancy is not: 955.6 observed against 494.8 ± 32.9 replicated ($p_{\text{ppc}} < 0.002$), and the observed maps are systematically *more compact* than the model predicts (mean maximum gap 4.469 against 4.688 ± 0.043). The residual co-harvest matrix (Fig. 10) shows exactly where it fails, and the failure is structured: because γ is fitted to the *total* adjacency, the model reproduces that total by construction but distributes it wrongly across the die—strongly over-predicting co-harvest at slices $(7, 8)$ and $(3, 4)$ (-4.8 and -3.7 standard deviations) while under-predicting it at $(7, 10)$ and $(0, 1)$ ($+4.0$, $+3.7$). **A single, position-independent adjacency term does not capture the spatial structure**; whatever produces the clustering is not uniform across the die. The term still captures a large part of it—the same discrepancy against the no-clustering model is 1631.9—so γ summarises a real effect rather than an artefact. We report the misspecification because a reader otherwise has no way to tell a fitted parameter from an adequate model.

K. Sensitivity to the analysis choices

Every number above depends on three discretionary choices: where to set the clean-day threshold, how to summarise a socket's clean days into one map, and how many draws to run the null for. We varied all three (Table IX).

TABLE IX
SENSITIVITY TO THE ANALYSIS CHOICES.

Choice	Sockets	Patt.	$P(s_0)$	t
clean-day threshold				
0.50	1,962	442	61.2 %	26.6
0.80	1,962	442	61.2 %	26.6
0.90	**1,962**	**443**	**61.2 %**	**26.6**
0.95	1,962	443	61.2 %	26.6
0.99	1,960	443	61.2 %	26.6
map rule, threshold fixed at 0.90				
mode	**1,962**	**443**	**61.2 %**	**26.6**
first clean day	1,962	442	61.6 %	28.4
last clean day	1,962	443	61.1 %	25.9
strict	1,756	436	60.8 %	28.0

Settings used in the paper are in bold. t is the Welch statistic for the lot contrast. Retained clean socket-days fall from 1,607,604 at threshold 0.50 to 1,324,982 at 0.99; *strict* keeps only sockets whose every clean day is identical.

The clean-day threshold is inert. Moving it from 0.50 to 0.99 discards 18 % of socket-days and changes nothing that matters: the same 1,962 sockets (1,960 at 0.99), 442–443 patterns, an identical 61.2 % slice-0 rate and an identical lot contrast. This is a stronger statement than "we chose 0.90".

The map rule matters slightly, and in the expected direction. Taking the first clean day instead of the mode changes 163 of 1,962 maps (8.3 %) and taking the last changes 49 (2.5 %) — these are exactly the sockets that undergo a transition, and the asymmetry (more change under "first" than "last") reflects that commissioning-era configurations are the ones later replaced. The lot contrast ranges only over

$t = 25.9$–28.4. The *strict* rule, which keeps only sockets whose every clean day is identical, retains 1,756 sockets — precisely the 89.5 % that never change — and still reproduces the boundary at $t = 28.0$.

The null is converged. Running the curveball null at 25, 50, 100, 200 and 400 draws gives $z = -18.3, -18.6, -17.9, -19.6, -20.0$: stable in magnitude with about ± 1 of Monte-Carlo noise, which is why Sec. IV-D quotes $z \approx -20$.

The thermal alignment grid was the one remaining untested choice; it has since been varied over 20 s–15 min and is reported in Sec. IV-E. It is the only choice we found that materially moves an effect size, which is why the affected number is now quoted at native resolution.

L. The map's effect on power, bounded rather than dismissed

Recovering a hidden hardware property is only useful if it changes something. We therefore ask whether the harvest map predicts socket behaviour, using a **within-node paired contrast**: the two sockets of a node share chassis, airflow, inlet air, power supply and—to first order—workload, so the difference $p_0 - p_1$ removes the node-level confounds that make cross-node power comparisons unreliable. We summarise each socket's map by the mean index of its active slices, the number of adjacent active pairs, and the count of active slices in the low half, and regress the socket-to-socket difference in each against the socket-to-socket difference in mean CPU power (Fig. 15).

Pairing is not by itself sufficient, as it does not remove confounding if load is distributed asymmetrically between the two sockets. We therefore also adjust for node utilisation, core frequency, inlet air temperature and month.

Failing to reject is not the same as showing negligible: An earlier version of this analysis used six months of power data, reported $r = +0.039$ (permutation $p = 0.003$) explaining 0.16 % of variance, and concluded that harvest maps are "operationally invisible". That conclusion does not survive two corrections, and we state the revision plainly because it changes a headline claim.

The first correction is the instrument. "No measurable effect" cannot be established by a small r; the question is whether the effect is small *enough*, which requires an equivalence test against a margin fixed in advance. We adopt ± 5 W per socket. Mean socket draw over this period is 69.0 W, so the margin is 7.2 % of it—deliberately generous in absolute terms, but still *tighter* than a difference operators demonstrably tolerate without modelling it: the p0–p1 positional asymmetry in the same data is 6.8 W. Anchoring the margin to revealed operator indifference is more defensible than picking a round number. The second correction is the uncertainty: node-days within a rack are not independent, and the previous interval treated them as if they were.

Re-run over the **full 31-month record (636,223 paired node-days on 980 nodes)**, adjusting for node utilisation, core frequency, inlet air temperature and month, with rack-clustered bootstrap intervals, the result is (Table X):

TABLE X
EQUIVALENCE TESTS AGAINST A ± 5 W MARGIN, COVARIATE-ADJUSTED, RACK-CLUSTERED.

Feature	Effect	90 % CI	TOST
mean slice index	$+7.9$	$[+0.9, +16.6]$	not equiv.
low-half count	$+5.8$	$[-2.2, +14.5]$	inconclusive
adjacent pairs	-1.4	$[-4.5, +1.7]$	**equivalent**

A 90 % interval inside the margin is the two-one-sided-tests criterion at $\alpha = 0.05$.

Only one of the three features can be declared negligible. For the mean active-slice index the interval excludes zero *and* extends to $+16.6$ W, so the data are consistent with an effect several times the margin. The point estimate is nonetheless modest in context, though not trivial: at 7.9 W it is 11 % of the 69.0 W mean socket draw, against a p0–p1 positional asymmetry of 6.8 W (10 %) and a socket-to-socket spread of 19.9 W (29 %). Position in the chassis matters about as much, and ordinary variation nearly three times more.

The honest summary is therefore *bounded but not negligible*: harvest-map variation moves socket power by an amount comparable to which slot a processor occupies, and we cannot rule out more. Schedulers that already ignore socket position may reasonably ignore this too, but the claim that the map is operationally invisible is withdrawn. Note also that this does not weaken the provenance argument of Sec. IV-G: causation still runs from manufacture to behaviour, never from rack position to a map fixed months before installation.

M. The map does not predict which dies get replaced

The map is a record of how a die came out of the fab; replacement is what the field eventually concluded about it. If dies whose harvested slices sit in a tight cluster were the ones that failed, that would tie manufacturing variability to operational reliability far more directly than any power correlation. We test it.

The design is a discrete-time monthly survival model over the post-commissioning window. A socket is at risk from its first steady-state clean day and censored at its last; the event is a CPU-SWAP or NODE-SWAP affecting that socket, with RELABEL excluded because no silicon moves. Covariates are taken from the socket's *first* steady-state map, never from one observed after the event, and comprise slice-0 status, the number of index-adjacent harvested pairs, the dispersion of the harvested indices, the rarity of the pattern ($-\log$ its fleet frequency) and the lot. This gives **92 sockets with at least one silicon-moving event over 53,597 socket-months at risk**, a monthly hazard of 0.172 %. (The 92 here and the 96 of Sec. IV-I count different things: 96 is dies moved, counting two per NODE-SWAP; 92 is sockets experiencing a first event.)

Nothing predicts anything. No coefficient's rack-clustered bootstrap interval excludes zero: per standard deviation the hazard ratios are 1.00 $[0.65, 1.51]$ for slice-0 status, 0.80 $[0.59, 1.02]$ for adjacent harvested pairs, 0.96 $[0.76, 1.23]$ for dispersion, 0.78 $[0.55, 1.06]$ for rarity and 0.84 $[0.61, 1.19]$ for lot. Under grouped seven-fold cross-validation over racks the out-of-fold concordance index is **0.448**, against a permutation null of 0.499 ± 0.031 ($p = 0.95$; Fig. 11, left)—no discrimination at all, and if anything below chance. Calibration is correspondingly flat: observed replacement rates across predicted-risk deciles range 0.08–0.26 % with no monotone trend.

We report this as a bounded null rather than an absence of evidence [OI-16]. The intervals above exclude hazard ratios beyond roughly ± 1.5 per standard deviation, so a large effect is ruled out even though a small one is not. Two readings are consistent with it: the defects that drive fusing at manufacture are not the ones that kill a part in the field, or replacement is dominated by causes unrelated to the die at all. Either way, an operator should not use the harvest map as a reliability covariate—which, with Sec. IV-L, completes a consistent picture of a property that is highly informative about *provenance* and uninformative about *behaviour*.

N. How much telemetry does the disclosure cost, and what closes it?

We have shown the maps are recoverable; an operator publishing such a dataset needs to know how cheaply, and which mitigations would work. Both follow from one asymmetry: a harvested core can never emit a sample, so the channel has *no false positives*. Recovery fails only if an active core stays silent for the whole window, so the question is not how long one must watch but how many samples per core survive collection.

Under healthy collection that question is almost vacuous. The median core reports **4,320 of 4,320** expected samples per day, so it appears in essentially every 20 s slot, and a **single sampling interval recovers the exact map for 99.70 % of sockets**. Measured directly and without any model, exact recovery holds for 99.9 % of sockets at *every* window from one day to the full month. Modelling the degraded regime, where each core yields λ samples in the window, exact recovery is $(1-e^{-\lambda})^{16}$: negligible at $\lambda \leq 1$, 0.44 at $\lambda = 3$, 0.90 at $\lambda = 5$ and 0.99 at $\lambda = 7$. The knee is around five to seven samples per core—**roughly two minutes of telemetry**.

That immediately rules out the obvious defences. Publishing at a coarser cadence does not work: recovery stays at 1.000 down to one sample per core per day, because a single sample names its core just as well as 4,320 do. Random dropout does not work either until it becomes destructive—recovery is still 1.000 after discarding 99.9 % of samples and only falls to 0.61 at 99.99 % (Fig. 11, right). **Reducing resolution is not a mitigation, because the channel is carried by which sensors *exist*, not by what they report.** Only defences that remove the per-core position work: publishing socket-level aggregates, or renumbering survivors compactly to $0 \ldots 15$—which is precisely what this BMC does not do, and the whole reason the channel exists. Per-node random renumbering destroys the map while still leaking the count.

V. Discussion

Why is slice 0 special?: The single most striking number is that slice 0 is harvested on 88.2% of Lot A sockets but only 39.3% of Lot B. Two explanations are live. Under *defect-driven* harvesting, fusing follows wherever killer defects land. Under *policy-driven* harvesting, slice 0 abuts a shared structure and is preferentially fused for reasons of yield policy, binning recipe or physical design; an 88% rate on one specific slice is very hard to obtain from random defects.

The fit of Sec. IV-J shows that *two statistically distinct effects are present*, and estimates them separately. The propensity vector β carries the slice-0 asymmetry and differs enormously between lots ($\beta_0 = +2.51$ versus a flat Lot B); the interaction parameter γ carries residual spatial association and is significantly positive in *both* lots. The most economical reading is a change of binning recipe or die source between the two procurements, superimposed on a spatial process present in both.

We are careful about what this does *not* establish. The decomposition is descriptive, not causal: a positive γ says harvested slices are spatially associated beyond what the per-slice propensities explain, and a correlated *defect* process and a binning *policy* that fuses contiguous blocks both predict exactly that. The model cannot tell them apart, and neither can any amount of data from one machine. Nor is the model adequate on its own terms (Sec. IV-J): it under-predicts how compact the observed maps are, so even the split between "propensity" and "clustering" is a summary of the structure rather than a complete account of it. Earlier drafts of this paper said the answer was "both"; that claimed more identification than the fit supports, and we withdraw it. What survives is that the two lots differ in *both* parameters, which is a statement about the populations, not about the mechanism.

The direction of causation is fixed.: A tempting objection to Sec. IV-G is that rack position is confounded with cooling and workload. It cannot be: harvesting is performed by fusing at manufacture, months before a part is installed, so nothing about a socket's position in the machine room can influence its harvest map. The only admissible causal direction runs from procurement batch to rack, which is precisely the provenance signal we claim.

How identifying is a harvest map?: Since the map is a per-die property, it is fair to ask whether it identifies a socket—which would raise a different set of questions from the provenance ones we pursue. It does not. Over 1,962 sockets the pattern distribution carries $H = 8.08$ bits, an effective $2^H = 270$ distinct maps against 495 possible (90.2 % of uniform), and only **4.9 % of sockets have a map unique in the fleet** (95 % CI $[4.0, 6.0]$); a typical map is shared by 4.4 sockets. Adding the lot raises this only to 12.1 %. The map is thus a *partial* identifier—strong evidence about which population a part came from, weak evidence about which part it is. That is precisely the claim we make, and it is the quantitative reason this work sits apart from the fingerprinting literature [13] rather than extending it.

What this buys an operator.: Three things. Fleet telemetry can audit a procurement without vendor cooperation, detecting whether a delivery is homogeneous—and, per Sec. IV-G, assigning an unseen rack to the right delivery 96 % of the time. It can detect field deconfiguration events that would otherwise be visible only in service logs. And it *bounds* how much a hardware property matters: the map does not predict replacement at all (Sec. IV-M), and its effect on power is comparable to which slot a processor occupies (Sec. IV-L). Bounding an effect is a weaker claim than dismissing one, and the right one to make here.

VI. Related work

All references in this section were verified against their original sources on 2026-08-05. Manufacturing variability in HPC has been studied mainly through its *performance* consequences—frequency and power spread under a power cap [14], turbo-boost variation across nominally identical processors [15], and accelerator-level variability in GPU-rich systems [16]. That literature treats the die as a black box characterised by its behaviour. Yield modelling, conversely, treats defect statistics directly: the negative-binomial family [9] and spatial wafer-map models [10] describe clustered killer defects but are validated on fab data unavailable to system owners.

A third, closer body of work concerns hardware provenance. Physical unclonable functions and inherent hardware identifiers exploit analog manufacturing variation to answer *which die is this*, for traceability and counterfeit detection [13]. That is a different question: we do not identify a die, we recover *what the vendor disabled on it*, without designed-in circuitry or physical access. Node-level anomaly and outlier detection in HPC [17], [18] likewise establishes *that* nodes differ without recovering the hardware structure responsible. The M100 ExaData release itself has an active downstream literature spanning telemetry foundation models, energy analytics and operational-data ontologies [19], [20], [21]; none of it examines per-core sensor presence.

We are not aware of prior work recovering per-die harvest maps from deployed-system telemetry, and that is the gap this paper addresses. We state this as an awareness claim rather than a priority claim: it rests on a systematic but necessarily incomplete search.

VII. Threats to validity and open issues

Every item below carries the identifier it is tracked under in `LIMITATIONS.md` in the artefact repository, which records for each one what is known, what would settle it, and where to look. Nothing here is load-bearing for a headline result unless the entry says so.

1) **Sensor presence $\neq$ core existence. Resolved externally** (Sec. III-B): over the whole record the operating system reports 128 logical CPUs on 99.9853 % of 847 M samples across 990 nodes (99.997 % of the 40.9 M samples in the single month 2022-08, across 988 nodes), and the scheduler's configured totals agree—all independently of the IPMI sensor family. The single OS-level anomaly in 2022-08 coincides with the only BMC-visible reconfiguration on that node. Supported internally by 99.9847 % of 1.67 M socket-days reporting exactly 16 cores, the perfect slice-pair structure, and the control of Sec. IV-H.

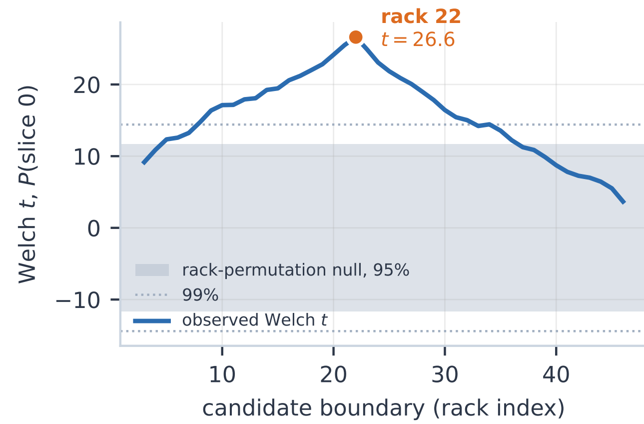

Fig. 9. The lot boundary against a simultaneous null. Welch t for $P(\text{slice }0)$ at every admissible boundary, with the envelope of $\max|t|$ under 4,000 rack permutations. The observed peak at rack 22 ($t = 26.6$) lies far outside a null that never exceeds 17.95, so the boundary is not an artefact of scanning many splits.

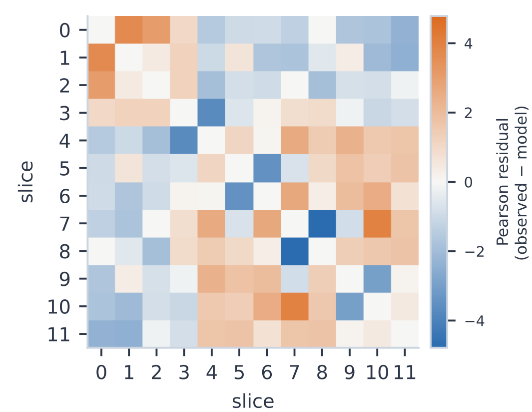

Fig. 10. Posterior-predictive residuals of the pairwise-interaction model: observed minus model-expected co-harvest counts, in standard deviations. The model is fitted to the *total* adjacency, so it matches that total but misplaces it—over-predicting at $(7, 8)$ and $(3, 4)$, under-predicting at $(7, 10)$ and $(0, 1)$. The clustering is real but not position-independent.

2) **[OI-1] Resolved.** The sensor index is recovered as the physical slice ordering from thermal coupling (Sec. IV-F), so the spatial reading of Sec. IV-D is supported. It remains an inference from thermal coupling rather than from a vendor floorplan; an annotated die shot would confirm it directly.
3) **[OI-2] Bounded.** A March 2020 BMC firmware update could have remapped sensor IDs, mimicking reconfiguration; 86 of the 235 transitions fall in the three days 2020-03-19–21. Those days lie inside the commissioning window excluded in Sec. IV-H, and every result is reported on the steady-state period, where the harvest maps are byte-identical and no comparable clustering occurs. Maintenance logs would still confirm the cause directly.

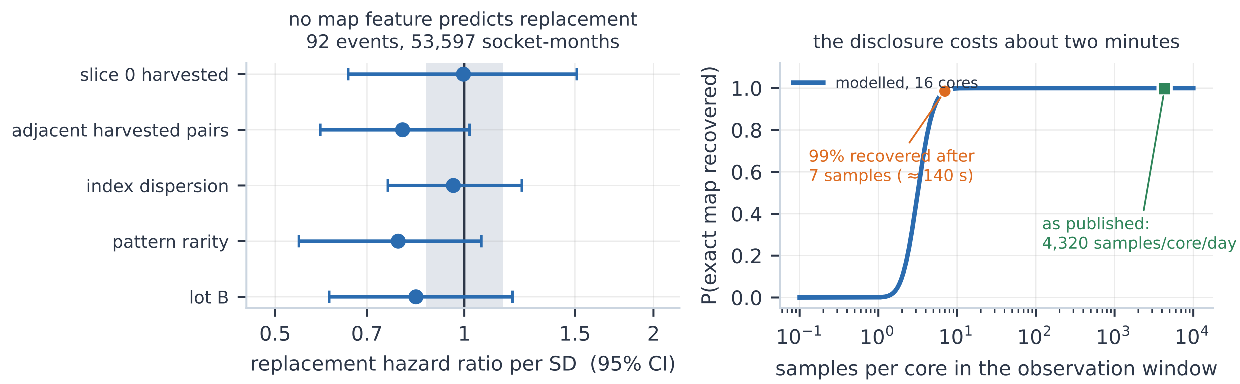

Fig. 11. Two bounded nulls. **Left:** no feature of the baseline harvest map predicts subsequent processor replacement; hazard ratios per standard deviation with rack-clustered bootstrap intervals, shading marking a $\pm 15\,\%$ negligible-effect band. Out-of-fold concordance is 0.448 against a permutation null of 0.499 ± 0.031. **Right:** the cost of the disclosure. Exact recovery needs about seven samples per core—roughly 140 s—while the published dataset ships 4,320 per core per day, three orders of magnitude beyond the knee. Reducing resolution is therefore not a mitigation.

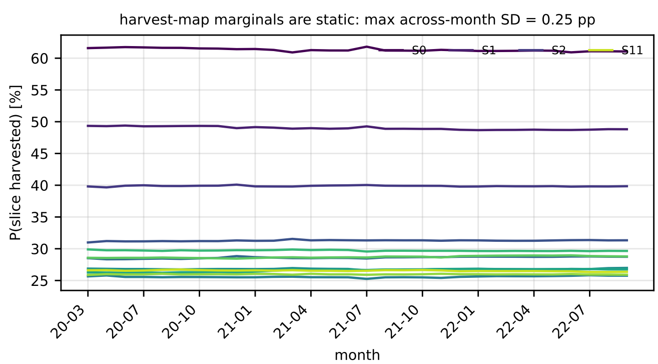

Fig. 12. Per-slice harvest rate in each of the 31 months. The largest across-month standard deviation of any slice is 0.25 percentage points.

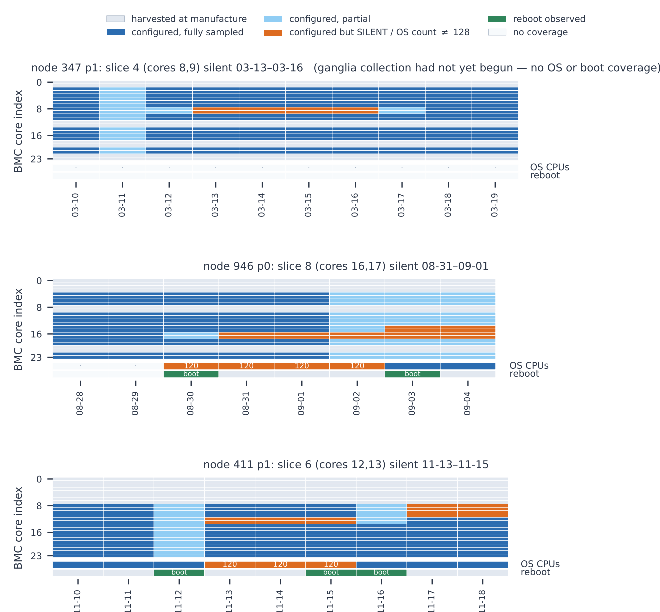

Fig. 13. The three core-guard episodes, with all three collectors that corroborate them. The grid is BMC per-core presence (grey = harvested at manufacture, blue = configured and reporting, orange = configured but silent); the strips below are the operating system's own logical-CPU count and the independent boot signal. In every case exactly one slice goes silent while all ten control sensors on the same node stay at full cadence; where coverage exists, the OS drops to $128 - 8 = 120$ over a window that brackets the BMC one, and the episode sits across observed reboots — what a GARD record applied by hostboot at IPL requires. Node 347 predates ganglia collection, so its strips are marked as absent coverage rather than left blank.

4) **[OI-4] Partly resolved.** All references have been verified against their original sources; two were corrected in the process (the negative-binomial model is Koren, Koren and Stapper in *IEEE T. Computers*, not Stapper alone in *T. Semiconductor Manufacturing*). What remains outstanding is the **novelty check**: our claim is an awareness claim, resting on a systematic but necessarily incomplete search.
5) **[OI-3] Defect-driven versus policy-driven harvesting is not identified.** A correlated defect process and a binning policy that fuses contiguous blocks both predict $\gamma > 0$, so the fit of Sec. IV-J measures residual spatial association without attributing it. An earlier draft claimed the decomposition showed "both"; that overstated what the model identifies and is withdrawn. The model is also demonstrably incomplete (Sec. IV-J): a single position-independent adjacency term reproduces the total adjacency it is fitted to but misplaces it across the die. A second installation, or vendor disclosure, would be needed to separate the mechanisms.
6) **[OI-5] Resolved.** Monthly coverage ranges from 18 % to 99.4 %, but the coverage filter does not drive any result: clean-day rates are near-identical between lots, coverage is uncorrelated with the harvest pattern ($r = +0.004$, $p = 0.88$), and restricting to $\geq 90\,\%$ coverage leaves the lot contrast unchanged (Sec. IV-G).
7) **[OI-6, OI-7] both resolved.** The sub-16 socket-days were dominated by an extraction defect on our side, now

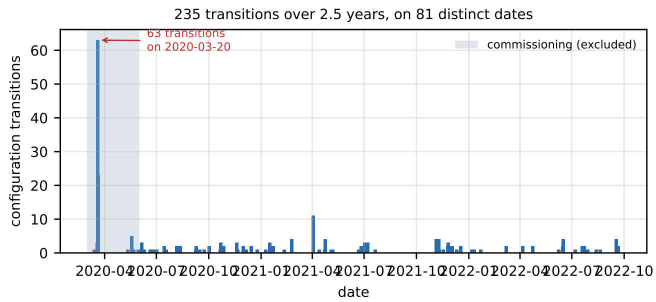

Fig. 14. Configuration transitions over the full record. 2020-03-20 alone accounts for 63, all inside the commissioning window (shaded) that Sec. IV-H excludes; the steady-state period shows no comparable cluster.

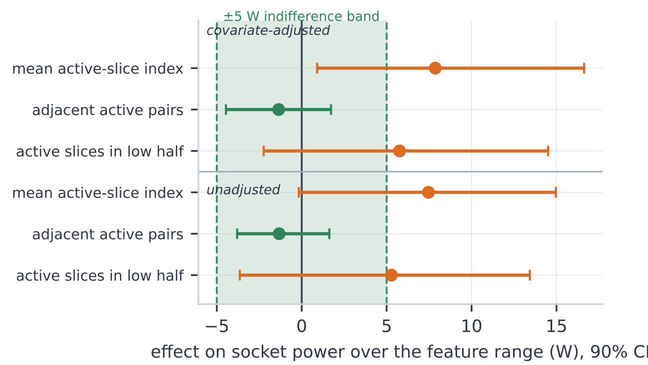

Fig. 15. Equivalence tests for the harvest map's effect on socket power, over 636,223 paired node-days. Points are the effect across each feature's full observed range with rack-clustered 90 % intervals; the band is the ± 5 W indifference margin. Only *adjacent active pairs* (green) is bounded inside it. The mean active-slice index is neither zero nor negligible, which is why the earlier "operationally invisible" claim is withdrawn.

gated; the nine that survive are the guard episodes of Sec. IV-H. The two nodes appearing to change within a day are same-day reboots.

8) **[OI-8] Largely settled.** Extending per-socket power collection from six months to all 31 raises the usable relabelling events from 4 to **23**. Restricting to events where the socket power difference is large enough for its sign to be meaningful (> 2 W on both sides), **12 of 15 flip sign** at the relabelling instant (80 %, 95 % CI $[52, 96]$). A collector-side tag swap carries every per-socket metric with it, and power overwhelmingly follows. The residual minority may be genuine die exchanges that happen to mirror, or power differences too small to resolve; any per-socket longitudinal study on this dataset should still treat relabelling as a hazard.

9) **[OI-12] The guard episodes rest on three events**, which is a strong claim from a small count. They are corroborated by three independent collectors—every other sensor on the socket kept full cadence, the OS flags the same three nodes and only those, and both steady-state episodes sit across observed reboots (Sec. IV-H). The third falls in the excluded commissioning window and predates `boottime` collection, so it is reported separately.

10) **[OI-15] Pair-level co-harvest counts are Monte-Carlo sensitive.** Under five seeds the Bonferroni-corrected test returns 16–18 significant pairs. The structure is invariant— exactly eight positive excesses at $|\Delta k| \leq 2$ every time—so we report the range and the invariant rather than one run's counts (Sec. IV-D).

11) **[OI-11] Common-mode artefact** in the thermal analysis, addressed by same-distance contrasts. Where per-pair samples were not retained, reported bands are spreads across core pairs rather than bootstraps over sockets, and the figures say which.

12) **Analysis choices moved our numbers more often than the data did.** We record this because it is the most transferable thing we learned. *Four* reported results changed materially during this work, none because of a new measurement.

    Sample size. A three-day sample overstated the within/cross-slice contrast threefold *and inverted the floorplan conclusion* (Sec. IV-F).

    Preprocessing. A one-minute alignment grid—adopted to work around an unrelated bug and never revisited—inflated that contrast by $2.2\times$ (Sec. IV-E).

    Bookkeeping. An intermediate revision quoted a corrected figure from a different subsample than the one it named, in two places.

    Uncertainty [OI-13]. Most consequentially, the claim that the map is operationally invisible rested on an interval that treated node-days within a rack as independent. Corrected, and extended from six months to all 31, the effect is neither zero nor bounded below the margin, and the claim was withdrawn (Sec. IV-L).

    Three of the four were introduced while fixing or extending something else, and every one of them ran in the direction we would have preferred. The defences that worked were mechanical rather than clever: re-deriving every number from a single stated pipeline, varying each discretionary choice (Sec. IV-K), clustering uncertainty at the level at which the data are actually dependent, and reporting the most conservative variant. Anyone mining a telemetry archive of this kind should expect the same failure mode, and should be at least as suspicious of their corrections as of their original analysis.

13) **[OI-10] Single system**; findings may reflect CINECA procurement rather than POWER9 generally.

VIII. Conclusion

Out-of-band telemetry that supercomputer operators already collect inadvertently publishes a manufacturing secret: which units a vendor fused off each individual die. Recovering these maps for 1,962 POWER9 sockets shows harvesting operates exactly at the slice—the unit sharing an L2/L3 block—with no exceptions in 1.67 M socket-days; that the maps are per-die rather than per-SKU, with 443 of 495 possible patterns realised; that harvested slices cluster *spatially*, the sensor index having been independently recovered as the physical ordering; and that a sharp changepoint at rack 22 separates two procurement populations with different binning statistics—sharply enough to assign an unseen rack to the right delivery 96 % of the time. Over 2.5 years the maps are static to within 0.25 percentage points per month, change at reboot and essentially nowhere else—95.5 % of transitions coincide with a boot observed by an independent collector—and exhibit three field core-guard episodes, two of them in steady state, each removing exactly one slice while every other sensor on the socket keeps reporting.

Two things deserve emphasis. The thermal probe recovers the physical slice ordering from telemetry alone, which is what licenses reading the clustering spatially and connects the measurement to the yield literature; it also shows how much this depends on sample size, since a three-day sample gave the opposite answer. And the harvest map's operational footprint is now *bounded* rather than dismissed: over 636,223 paired node-days it moves socket power about as much as which slot a processor occupies, and it does not predict replacement at all—bounded nulls we consider as useful as the positive ones, since they tell operators what they can safely ignore and what they cannot yet. The broader point is that fleet telemetry is a viable instrument for hardware provenance, and that its unintended disclosures deserve attention from those who publish such datasets.

Artefact availability

All analysis code, the derived per-(node, socket, core, day) tables, and the recovered harvest maps are available at the artefact repository accompanying this paper, together with `LIMITATIONS.md`, the register the identifiers in Sec. VII refer to. Every result is reproducible from the public M100 ExaData Zenodo releases [1]. The derived tables (15 MB) suffice for all structural and longitudinal results without re-downloading the 49.9 TB source. Two sections need one further pass over the archives, and the script that performs it is included: the thermal analysis reads the raw per-core series, and the reboot and power sections read a per-(node, day) summary of `boottime`, `cpu_speed`, `cpu_user`, `load_one` and `ambient`.

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